

MESSAGE AUTHENTICATION REQUIREMENTS

Message authentication is a mechanism or service used to verify the integrity of a message. Message authentication assures that data received are exactly as sent by the sender without any modifications, insertions, deletions or replay and that the sender identity is valid.

The following are various requirements for message authentication in the context of communication across a network:

- **Disclosure:** release of message contents to any person or process not possessing the appropriate cryptographic key.
- **Traffic analysis:** discovery of the pattern of traffic between parties. In a connection – oriented application, the frequency and duration of connections could be determined. In connection – oriented or connection less environment, the number and length of messages between parties could be determined.
- **Masquerade:** insertion of messages into the network from a fraudulent source. This includes the creation of messages by an opponent that are claimed to come from an authorized entity. Fraudulent acknowledgements of message receipt or no receipt by someone other than message recipient.
- **Content modification:** changes the content of the message including insertion, deletion, transposition, and modification.
- **Sequence modification:** any modification to a sequence of messages between parties, including insertion, deletion and reordering.
- **Timing modification:** delay or replay of messages. Entire session or sequence of messages could be a replay of previous valid session or individual messages in the sequence could be delayed or replayed.
- **Source repudiation:** denial of transmission of message by source.
- **Destination repudiation:** denial of receipt of message by destination.

The first two attacks can be dealt with message confidentiality. Items 3 through 6 can be dealt with message authentication, for item 7 can be handled using digital signatures, and item 8 will require combination of digital signatures and a protocol designed to counter this attack.

Message authentication is a procedure to verify that received messages come from intended source and are not altered. It can also verify sequencing and timeliness.

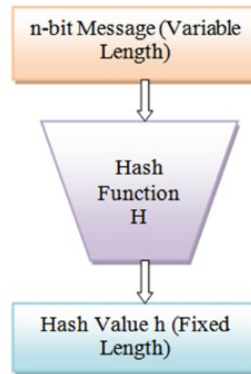
MESSAGE AUTHENTICATION FUNCTIONS

Any message authentication and digital signature mechanism has two levels of functionality. At the lower level, there must be some sort of function that produces authenticator; which is a value to be used for authentication of message. This lower level function is then used as a primitive higher level authentication protocol that enables a receiver to verify the authenticity of a message.

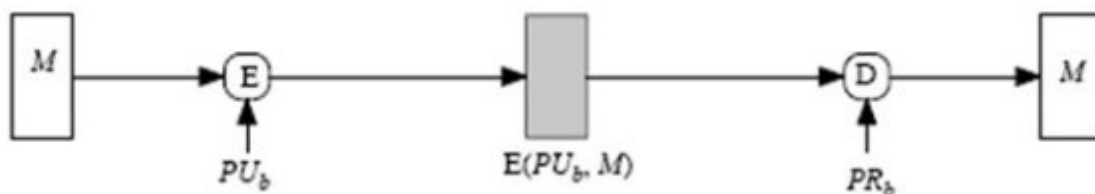
There are three types of functions that can produce a message authenticator:

- Hash functions
- Message encryption
- Message Authentication Code(MAC)

Hash functions: a hash function H accepts a variable – length block of data M as input and produces a fixed – size hash value $h = H(M)$. A hash function maps a variable – length message into a fixed – length hash value called as message digest. The following diagram demonstrates a hash function:



Message Encryption: message encryption by itself can provide a measure of authentication. It can be implemented both in symmetric and asymmetric encryptions. The following diagram demonstrates use of public key encryption, which provides confidentiality but not authentication.



The source A uses public key PU_b of destination B to encrypt M . Because B only has the corresponding private key PR_b , only B can decrypt the message. Authentication is not provided because any opponent can use B's public key to encrypt a message and claim to be sender A.

To provide authentication, A uses its private key to encrypt the message, and B uses A's public key to decrypt as shown below:



This provides authentication as the message must have come from A because A is the only party that possesses PR_a .

To provide confidentiality and authentication, A can encrypt M first using its private key, which provides the digital signature, and then using B's public key, which provides confidentiality as shown below:



Message Authentication Code: an alternative authentication technique involves the use of a secret key to generate a small fixed size block of data, known as cryptographic checksum or MAC that is appended to the message. This technique assumes that two communicating parties, say A and B share common key K. when A has a message to send to B, it calculates the MAC as a function of the message and the key:

$$MAC = MAC(K, M)$$

Where

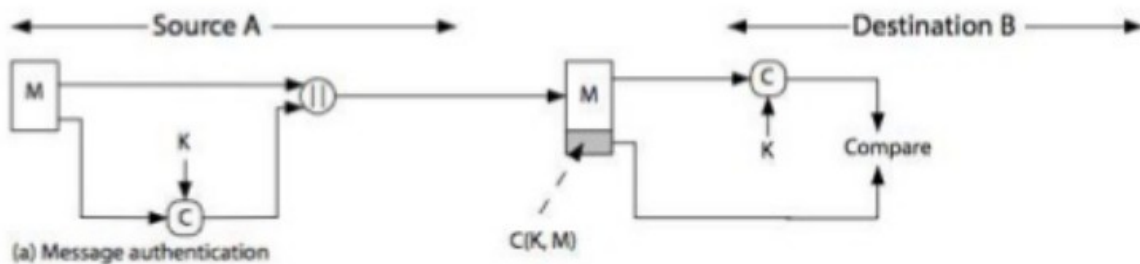
M = input message

MAC() = MAC function

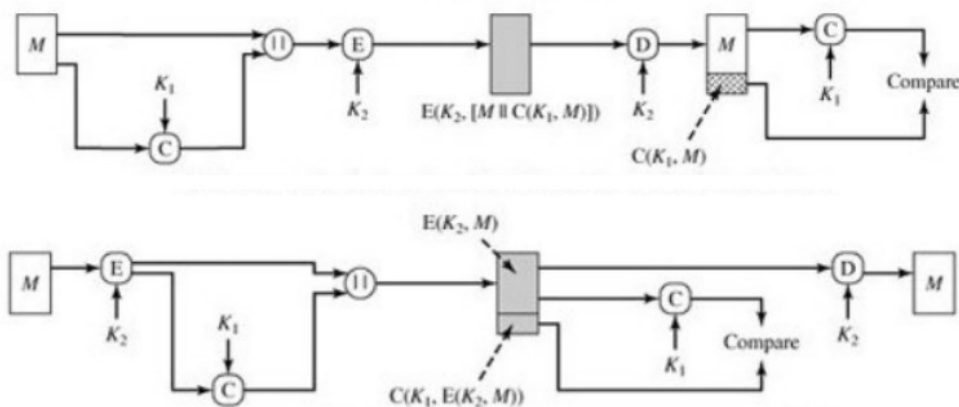
K = shared secret key

MAC = message authentication code.

The message and MAC are transmitted to intended recipient. The recipient performs same calculation on the received message, using the same secret key, to generate a new MAC. The received MAC is compared to the calculated MAC.



The above method provides authentication but not confidentiality. It can be provide by message encryption either after or before MAC algorithm:



In the first case MAC is calculated with the message as input and is then concatenated to the message, the entire block is then encrypted. In second case, the message is encrypted first then the MAC is calculated using the resulting cipher text and is concatenated to the cipher text to form the transmission block.
